

Beyond Calories: A Multimodal Deep Learning Comparison for Nutritional Health Prediction

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Abstract

Photo-based nutrition tracking is a real consumer problem that apps like CalAI and MyFitnessPal already address, but their model architectures are private. This project compares three open deep learning architectures on a nutrition prediction task to see which one works best. Using the Nutrition5k dataset from Google Research, the goal is to predict four nutrition values (calories, fat, carbs, protein) and a 3-class health label (healthy, moderate, indulgent) from a food image and an ingredient text list. The three models are: a fully connected baseline (FCNN), a CNN-only model using ResNet50 pretrained on ImageNet, and a multimodal fusion model that combines ResNet50 with DistilBERT for text and a small MLP for synthetic lifestyle features. The multimodal model wins on most metrics: 65.3% test accuracy compared to 61.1% (CNN) and 48.6% (FCNN). The pattern shows that pretrained encoders and combining modalities both help, but one finding (carbs) shows that adding a redundant modality can also hurt.

Keywords

multimodal learning, transfer learning, food computing, regression, classification

1 Introduction

Most diet tracking apps focus on calorie counts. But two meals at the same calorie count can have very different effects on the body depending on what is actually in them. A 600 kcal salad and a 600 kcal plate of fries are not the same.

This project takes a food image and a list of ingredients as input, and predicts both the nutrition values and a 3-class health rating. The goal is not to invent a new model. Instead it is to compare three architectures from different deep learning families on the same task with the same data, so I can see which one works best and why.

Contributions.

- A controlled three-way comparison of FCNN, CNN, and multimodal fusion models on Nutrition5k under the same data splits and same output heads.
- Evidence that pretraining (using ImageNet weights) gives the biggest improvement on small food image datasets.
- A clear case where adding a redundant modality (text describing visible carbs) can actually hurt performance.
- All code in a single Jupyter notebook using TensorFlow/Keras.

The rest of the paper is organized like this. Section 2 defines the problem. Section 3 describes the three model architectures. Section 4 reports the experiments and results. Section 5 covers related work. Section 6 concludes.

2 Problem Formulation

Inputs. For each dish I have:

- An overhead RGB image, sized 224x224x3
- An ingredient list as a comma-separated text string
- A synthetic lifestyle profile with 8 features (age, BMI, exercise, sleep, etc.)

Outputs. The model produces 7 numbers per dish:

- 4 regression values: calories, fat (g), carb (g), protein (g)
- 3 probabilities: P(healthy), P(moderate), P(indulgent), summing to 1

Targets. The nutrition values come straight from Nutrition5k, where each dish's calories and macros are calculated from per-ingredient nutrition in the USDA FoodData Central database. The health label is not in the dataset, so I made my own using rule-based thresholds (described in Section 3.5).

Loss. The model is trained with mean squared error on the standardized regression targets plus a 3-class cross-entropy loss on the health classification. The classification loss is weighted at 0.5 so it doesn't dominate the regression loss.

3 System and Algorithm Design

3.1 Overall System

There are three models. All three produce the same 7-dimensional output through the same output heads. Only the encoder part changes between them. This way any difference in performance is because of the encoder, not the prediction layer.

Table 1: Architecture comparison. All three models share the same output heads.

Model	Inputs	Encoder
FCNN	flattened image + ingredient bag-of-words	3 dense layers with dropout
CNN	food image only	ResNet50 (frozen) + dense head
Multimodal	image + text + lifestyle	ResNet50 + DistilBERT + small MLP, fused by concatenation

3.2 FCNN Baseline

The FCNN is the simplest model and uses no transfer learning. It treats inputs as one big flat vector.

The image gets shrunk from 224x224x3 down to 32x32x3 by average pooling and then flattened into a single vector. The ingredient text is encoded as a bag-of-words count vector (200 features) using

only training set vocabulary. The two are concatenated into one input vector and passed through three dense layers (256, 128, 64 units) with ReLU activation and dropout. The same output heads then produce the 7-dim output.

The image is heavily downsampled because at full resolution the input would have over 150,000 features, which is too much for a simple FCNN to learn from only 332 training examples.

3.3 CNN Encoder

The CNN model uses only the food image. The image goes through ResNet50, which was pretrained on ImageNet. ResNet50 outputs a 2048-dim feature vector after global average pooling. I then project this to 128 dims with a dense layer and pass it to the same output heads.

The ResNet50 weights are frozen, meaning I don't retrain them. I only train the small projection layer and output heads. With only 332 training dishes, fine-tuning all 23 million parameters in ResNet50 would overfit immediately.

This model tells us how much I can learn from just looking at the picture, with no text or lifestyle info.

3.4 Multimodal Fusion

The multimodal model has three input branches:

Image branch. Same as the CNN model: ResNet50 frozen, projected to 128 dims.

Text branch. The ingredient text is tokenized with the DistilBERT WordPiece tokenizer to 32 tokens. The tokens go through DistilBERT (also frozen). I take the [CLS] token output (768 dims) and project it to 128 dims with a dense layer.

Lifestyle branch. The 8-dim lifestyle vector goes through a small two-layer MLP with 32 units each, output is 32 dims.

The three branches are concatenated into one 288-dim vector, passed through a fusion dense layer back to 128 dims, then to the output heads. This is the model that should benefit the most from combining modalities.

3.5 Health Level Label

Nutrition5k doesn't have a health label, so I made one up. A meal is labeled *healthy* if calories are below 500, fat ratio is below 40%, and protein ratio is above 15%. It is labeled *indulgent* if calories are above 700 or fat ratio is above 50%. Otherwise it is *moderate*. These thresholds are picked to roughly balance the three classes. This is a rule-based label, not a clinical one, made for this assignment.

4 Experimental Evaluation

4.1 Methodology

Data. Nutrition5k contains about 5,000 cafeteria dishes from Google offices. Each dish has an overhead RGB image, an ingredient list with masses, and ground truth nutrition values from USDA Food-Data Central. For this project I used a sample of 500 dishes because google colab wanted me to pay when I tried to scale up to 3300 image on a 75 15 15. After filtering empty plates (mass below 30g) and outliers above 2,500 calories, 475 dishes remained.

The lifestyle features are synthetic. They are generated as random plausible values for age, BMI, exercise frequency, sleep, water

intake, calorie target, dietary restriction flag, and activity level. I use them as a placeholder to demonstrate the multimodal architecture supports a third modality.

Splits. 70/15/15 train/validation/test split, stratified by health label so all three sets have the same class proportions. That works out to 332 train, 71 validation, 72 test.

Preprocessing. Images resized to 224x224 and processed with the standard ResNet50 mean subtraction. Ingredient text is lower-cased and either bag-of-words encoded (for FCNN) or tokenized for DistilBERT (for multimodal). Regression targets and lifestyle features are standardized using train set statistics only.

Training. All models trained on a Colab T4 GPU using Adam optimizer with learning rate 1e-3. Batch size 32 (FCNN, CNN) or 16 (multimodal). Pretrained ResNet50 and DistilBERT weights are frozen. Training time was 12.5 seconds for FCNN (30 epochs), 59 seconds for CNN (25 epochs), and 84 seconds for multimodal (25 epochs).

Methods Compared. The three models from Section 3.

Metrics. For regression: per-target RMSE, MAE, and R^2 . For classification: accuracy, macro-F1, weighted-F1, per-class precision and recall, confusion matrix, and one-vs-rest ROC AUC.

4.2 Results

4.2.1 Training Behavior. Figure 1 shows validation loss and validation classification accuracy across training. All three models converged. The multimodal model reached the lowest validation loss and highest validation accuracy. The FCNN baseline plateaus around 40% validation accuracy while the two pretrained-encoder models keep improving.

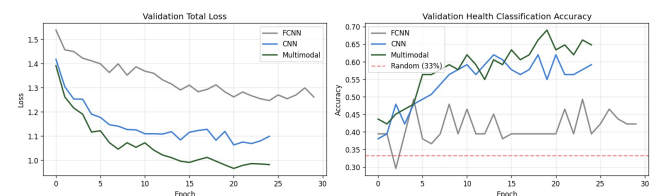


Figure 1: Validation loss (left) and validation accuracy (right) across training epochs. Multimodal < CNN < FCNN on loss; reverse on accuracy.

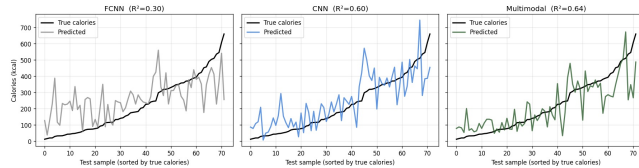
4.2.2 Regression Performance. Table 2 shows the R^2 values per nutrient on the test set. The multimodal model wins on three of four nutrients. Calorie R^2 goes from 0.30 (FCNN) to 0.60 (CNN) to 0.64 (Multimodal). Mean absolute error on calories drops from 108 kcal to 86 to 74. Fat and protein follow the same pattern. Carbs are the exception: CNN beats Multimodal at 0.32 vs 0.23.

Figure 2 shows a lifted chart for calorie predictions. The black line is true calories sorted in ascending order, and the colored line is the model's prediction for those same dishes in the same order. Multimodal tracks the truth most closely, especially at the high-calorie end.

4.2.3 Classification Performance. Table 3 shows the health classification metrics. Multimodal wins on all three measures.

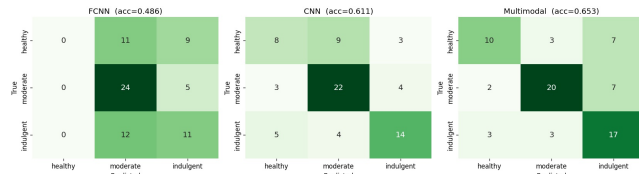
Table 2: Regression R^2 per nutrient on the test set. Higher is better.

Model	Calories	Fat	Carb	Protein
FCNN	0.30	0.28	-0.00	0.23
CNN	0.60	0.54	0.32	0.55
Multimodal	0.64	0.58	0.23	0.59

**Figure 2: Lifted chart for calorie predictions on test set.****Table 3: Health-level classification metrics on test set.**

Model	Accuracy	Macro F1	Weighted F1
FCNN	0.486	0.363	0.401
CNN	0.611	0.589	0.604
Multimodal	0.653	0.643	0.653

The confusion matrices in Figure 3 show something interesting. The FCNN never predicted the healthy class at all (zero correct out of 20). It just defaulted to the majority class. CNN started actually predicting healthy correctly (8 out of 20). Multimodal got 10 out of 20 with fewer cross-class errors.

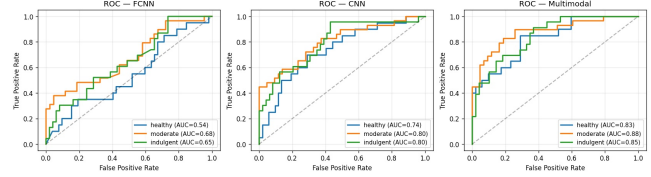
**Figure 3: Confusion matrices on test set. FCNN never predicts healthy; CNN and Multimodal both have a balanced diagonal.**

The ROC curves (Figure 4) confirm the same ordering. AUC for the healthy class jumps from 0.54 (basically random) for FCNN to 0.83 for Multimodal.

4.2.4 Discussion. Two main findings.

First, pretraining is the single biggest improvement over the baseline. The CNN-only model gains 12.5 percentage points of accuracy and roughly doubles calorie R^2 over FCNN, just by using a frozen ResNet50 backbone instead of training dense layers from scratch. Transfer learning from ImageNet works on food images even without food-specific pretraining.

Second, ingredient text and lifestyle add a smaller but consistent gain on top of the CNN. Multimodal gets 4.2 more points of accuracy and 0.04 better calorie R^2 than CNN-only. The biggest improvement

**Figure 4: ROC curves per class (one-vs-rest). AUCs go from 0.54-0.68 (FCNN) to 0.74-0.80 (CNN) to 0.83-0.88 (Multimodal).**

is on the minority healthy class, where per-class F1 goes from 0.44 to 0.57.

The carb finding is the most interesting. CNN actually beats Multimodal on carb R^2 (0.32 vs 0.23). Many high-carb foods (rice, pasta, bread, potatoes) are very visually distinctive, so the text saying “rice” adds nothing the image didn’t already show. When a modality is redundant rather than complementary, fusion can hurt by adding noise.

About overfitting: training accuracy reached 87% (CNN) and 89% (Multimodal) but validation stayed in the 60s. That’s overfitting from the small training set. To fix this I’d need data augmentation, early stopping, or just more data.

5 Related Work

Image-based nutrition. Thames et al. [1] introduced Nutrition5k and showed CNN regression can match professional nutritionists on calorie estimation. Their problem and method are very close to ours. I differed by adding text as a second modality and explicitly comparing across model families instead of just reporting one best model.

Recipe generation. Salvador et al. [2] did Inverse Cooking, which predicts ingredient lists and recipes from food images using a CNN encoder paired with a Transformer decoder. Their problem is generation, ours is regression and classification. I borrowed the idea of combining vision and text.

Depth fusion. Han et al. [3] did DPF-Nutrition, which fuses RGB with predicted depth maps to improve nutrition estimation on Nutrition5k. Their second modality is depth. Ours is text. Text is cheaper to acquire because it doesn’t need special hardware.

Architectures. Our visual encoder is ResNet50 [4]. Our text encoder is DistilBERT [5], a smaller faster version of BERT [6]. Both are pretrained on large general-domain datasets.

Earlier work. Im2Calories [7] from 2015 was an early attempt at calorie estimation from photos. They used object detection plus volume reconstruction, not end-to-end deep learning. Modern pre-trained encoders make end-to-end regression way easier and more accurate.

6 Conclusion

This project compared three deep learning architectures on a nutrition prediction task. The expected ordering held: a multimodal fusion of pretrained CNN, pretrained Transformer, and lifestyle MLP outperformed CNN-only, which outperformed an FCNN baseline. The multimodal model got 65.3% test accuracy on health classification and 0.64 calorie R^2 , and was the best on every metric except carb R^2 . The carb exception is itself useful information: it tells us

that text-based ingredient descriptions are most useful when they describe things not already visible in the image.

The pattern is common in multimodal deep learning. Pretraining is the biggest single win. Adding modalities helps but only when they actually contribute new information.

7 Work Division

This was a solo project. All work, including dataset acquisition, preprocessing, model implementation, training, evaluation, plot generation, and writing, was performed by the author. **AI tools (Claude by Anthropic) were used heavily for code scaffolding and report drafting.** All design and analysis decisions, all experimental runs, and final review were done by the author.

8 Learning Experience

This project taught me a lot about how a real deep learning research workflow actually goes. Three things stood out.

First, fair comparison is harder than it looks. The output heads have to be identical across models, otherwise you can't tell whether the difference is the encoder or just the head. This sounds obvious but I almost didn't get it right at first.

Second, pretraining matters more than I expected. The from-scratch FCNN baseline barely beat random on the minority healthy class. The pretrained CNN was way better even though I only

trained the head. That single change (ImageNet weights) gave a bigger improvement than anything else.

Third, multimodal gains depend on whether the modalities actually give different information. The carb result was unexpected at first but makes total sense in hindsight. If text just describes what's already obvious in the image, fusion can make things worse, not better.

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